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RESEARCH INTEREST

Computer Vision, Embodied AI, World Models.

Compositional Generative Model, Scene Understanding, Spatial Reasoning.

EDUCATION

ShanghaiTech University

B.E. in Computer Science and Technology, Minor in Life Sciences

Shanghai, CHN

September 2023 - June 2027 (expected)

– AI Honor Class

– Overall GPA: 3.83/4.0 rank 9/173 in CS major

– General Evaluation Ranking: rank 1/173 in CS major

– Relevant Coursework: Introduction to Information Science and Technology(A+), Introduction to Programming(A), Algorithms and Data Structures(A), Introduction to Machine Learning(A-), Artificial Intelligence in Medical Imaging(A), Computational Science and Engineering(A+), Computer Architecture(A-) & Project(A), Protein Design(A+), Game Theory(A)

Harvard University

Visiting Undergraduate Student

Cambridge, USA

September 2025 - May 2026

– Overall GPA: 4.0/4.0

– Relevant Coursework: Signal Processing (MIT cross-registration)(A), Computer Vision(A), Planning and Learning Methods in AI(A), AI for Molecular Biology(A), High Performance Computing(A-)

AWARDS & HONORS

○ **MERIT STUDENT**(Top 2% in school), ShanghaiTech University, 2023-2024

○ **MERIT STUDENT**(Top 1 in CS major), ShanghaiTech University, 2024-2025

○ **AI HONOR CLASS**(Honors Degrees), ShanghaiTech University, 2024-2027(expected)

○ **GOLD MEDAL**, International Genetically Engineered Machine Competition (iGEM), 2024

○ **FIRST PLACE**, Analytical Performance, International Student Competition on Biosensors ([SensUs](#)), 2025

○ **2023 OUTSTANDING MENTOR ASSISTANT**, ShanghaiTech University, 2023

RESEARCH EXPERIENCE

Perception, Learning and UnderStanding (PLUS) Lab, ShanghaiTech

Jan. 2025 - Sep. 2025

Undergraduate Research Assistant, Supervisor: [Prof. Xuming He](#)

○ Investigated **Compositional Scene Generation** with Scene Graph based Diffusion Model.

○ Focused on learning the disentangled representation from the scene graph, enabling flexible control over the generation process.

○ Explored Classifier-Free Guidance and Training-Free Method in Image Generation.

Embodied Minds Lab, Harvard University & Kempner Institute

Sep. 2025 - present

Visiting Undergraduate Research Assistant, Supervisor: [Prof. Yilun Du](#) & [Dr. Ruojin Cai](#)

○ Developed an **inverse generative modeling** framework for scene understanding on both synthetic and real-world datasets, training a relation-conditioned composable diffusion model that generates scenes from structured object-attribute and spatial-relation inputs.

○ Focused on unsupervised **object-relation discovery with Diffusion Model**, enabling generative models to perform text-image attribution analysis on both synthesized and real-world images.

○ Investigated the potential of improving compositional generation with unsupervised relation discovery in a training free manner, achieve **SOTA** performance on 2D spatial relation dataset VISOR.

SELECTED COMPETITIONS

PACIFY - iGEM 2024 [\[wiki\]](#)

Dec. 2023 - Oct. 2024

Team Member

- Performed homology modeling to obtain the structure of $\beta 10 - E5 - \beta 11 - K5$, and used AlphaFold 2 to predict the structure of $\beta 1 - \beta 9$
- Operated protein preparation and molecular dynamics simulation
- Developed devices based on PID algorithm to address the issue of itchiness without doing harm to the skin

MakeSense, ShanghaiTech First SensUs Team

Aug. 2024 - Aug. 2025

Co-Founder & Leader of Data Analysis Team

- Developed a wearable device based on biosensors to continuously monitor acute kidney injury (AKI) biomarkers.
- Invested in an enzyme-based creatinine sensor and QCM (Quartz Crystal Microbalance) platform.
- Set up a data analysis pipeline to process the sensor data, achieving near-perfect accuracy in predicting creatinine concentration.
- **First Place** in Analytical Performance: "This team did an absolutely remarkable job with unprecedented results and near-perfect accuracy. This is a first in SensUs history and sets a new benchmark for other teams, especially as a first-time participating team." by SensUs Committee.

SELECTED PROJECTS

De Novo Design of Odorant Binding Proteins for Breast Cancer Detection

Dec. 2024 - Jan. 2025

Course Project, Supervisor: Prof. Jiayi Dou

- Designed three novel Odorant Binding Proteins (OBPs) to specifically recognize Volatile Organic Compounds (VOCs)—hexanal, octanal, and nonanal—that serve as biomarkers for breast cancer.
- Executed a complete de novo computational design pipeline, generating protein backbones with RFdiffusionAA and designing amino acid sequences using LigandMPNN.
- Validated designs using AutoDock, PyRosetta, and ESMFold, demonstrating that the engineered proteins achieved significantly higher binding affinity and stability compared to their natural counterparts.

Neural Olfactory Sensing and Evaluation (NOSE)

May. 2025 - June. 2025

Course Project for Introduction to Machine Learning, Supervisor: Prof. Yujiao Shi

- Fine-tuned **MolFormer**, a large chemical language model, on the GS-LF olfactory dataset to adapt its generalized representations for specialized odor prediction tasks.
- Evaluated the model against the state-of-the-art **OpenPOM** on the Keller-2016 dataset, performing odor label classification and pleasantness rating prediction.
- Achieved state-of-the-art performance, with the fine-tuned model matching or surpassing OpenPOM on key metrics, demonstrating the efficacy of **fine-tuning for machine olfaction**.

Phase-Adaptive Quantization for AI Accelerators

Apr. 2026 - May. 2026

Course Project for Hardware Architecture for Deep Learning, Supervisor: Prof. Joel Emer & Prof. Vivienne Sze

- Designed a **phase-aware 4-bit quantization design-space exploration** framework for LLM, VLM, and VLA inference workloads, comparing prefill and decode regimes across NVFP4-like, MXFP4-like, and custom rescale pipelines.
- Built an automated **AccelForge** experiment flow to generate workload/architecture YAMLS, run hardware energy/latency sweeps, extract per-einsum bottleneck breakdowns, and resume long-running Docker-based mappings.
- Implemented a Python quantization accuracy emulator using real model tensor snapshots to evaluate per-layer cosine similarity and combine accuracy with hardware cost for Pareto frontier analysis.
- Demonstrated that decode and prefill favor different quantization configurations, motivating **phase-adaptive datapaths** over a single fixed 4-bit format.

TECHNICAL STRENGTHS

Programming Languages

Matlab, Python, C&C++, RISC-V Assembly

Framework & Toolchain

PyTorch, Git, Docker, Linux, Rosetta, CUDA, AccelForge, OpenMP, MPI

Misc

L^AT_EX, Markdown, IELTS 7.5(6.5), CET-6(646)